

KnowSelf: Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Enabling agents to autonomously regulate knowledge utilization through situational self-awareness

Source: arxiv.org/abs/2504.03553

Current Agent Training Uses 'Flood Irrigation'

Indiscriminate Knowledge Injection Hurts Efficiency

Current LLM agent methods resemble unconscious pattern-fitting — they blindly inject knowledge regardless of whether the agent actually needs it at each step.

1. Trajectory Imitation

ReAct-style methods copy expert trajectories without learning situational context

2. Trial-and-Error Refinement

Reflexion-style methods rely on external feedback for self-correction

3. Knowledge-Augmented

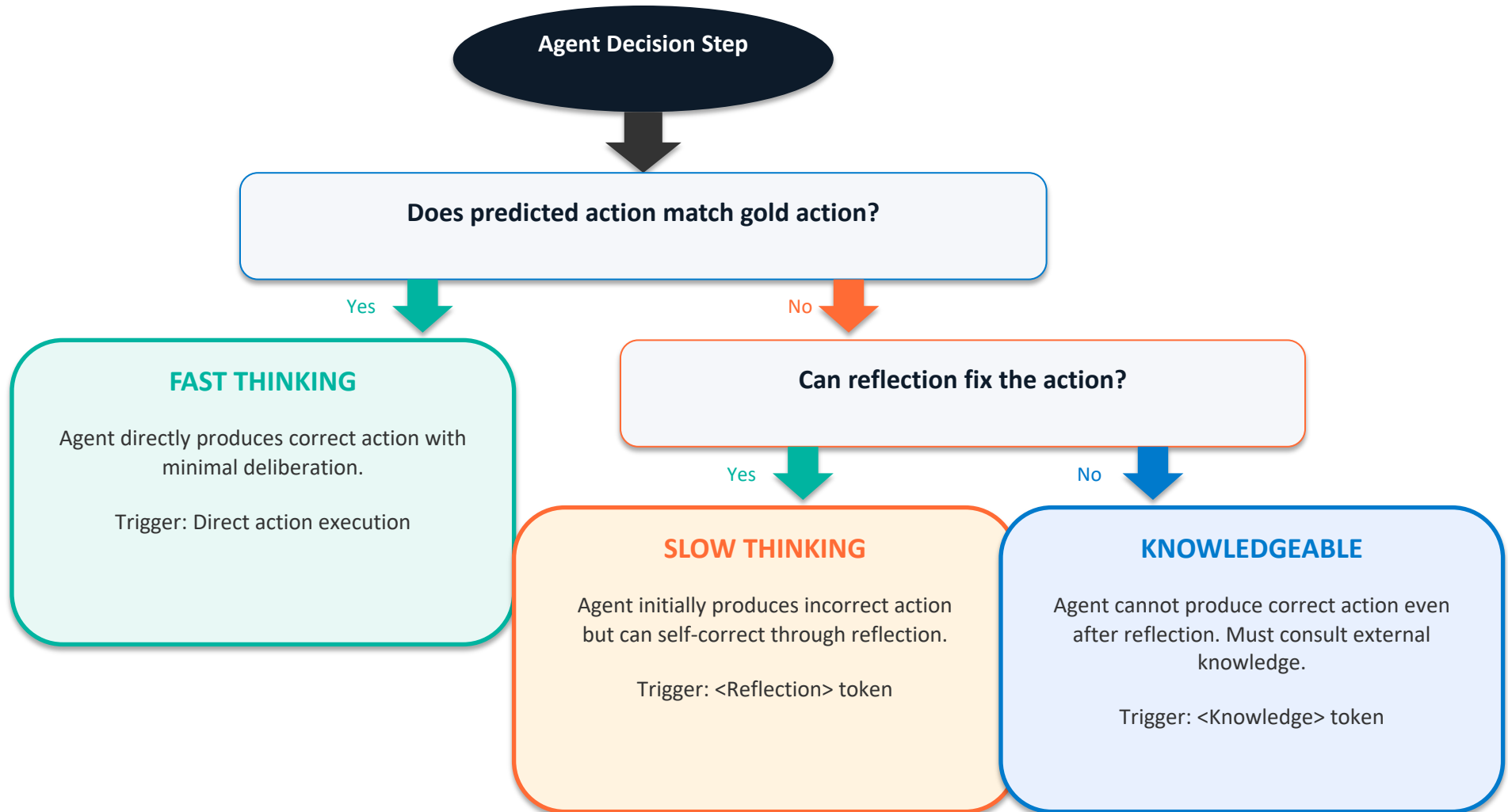
KnowAgent/WKM methods retrieve external knowledge for every query

Key Issue: All three approaches use "flood irrigation" — applying the same knowledge strategy to every situation regardless of actual need. This leads to pattern collapse, fragile planning, and unnecessary inference costs.

Humans possess situational self-awareness: the metacognitive ability to assess whether we can handle a situation directly, need to reflect, or must seek help. KnowSelf brings this capability to LLM agents.

Three Situational Modes: Fast, Slow, and Knowledgeable Thinking

KnowSelf identifies three cognitive situations based on the agent's ability at each decision step



Special tokens (<Fast>, <Reflection>, <Knowledge>) are inserted during training to mark each situation type.

KnowSelf Framework: Three-Stage Pipeline

Data Construction -> Two-Stage Training -> Self-Aware Inference

1

Self-Awareness Data Construction

Agent self-explores environments
Heuristic classifies each step:
- Action matches gold -> Fast
- Reflection fixes -> Slow
- Neither works -> Knowledge
Special tokens mark situations

2

Two-Stage Training

Stage 1: Supervised Fine-Tuning (SFT)
-> Teaches initial self-awareness
Stage 2: Reward-based Policy Opt (RPO)
-> Refines token generation
-> Learns appropriate triggering

3

Self-Aware Inference

Agent generates special tokens
<Fast> -> Direct action execution
<Reflection> -> Self-correction loop
<Knowledge> -> External KB retrieval
Selective, adaptive response

Framework illustrated in Figure 3 of the original paper (arxiv.org/abs/2504.03553)

KnowSelf Correctly Handles Tasks Where Baselines Fail

Qualitative comparison of KnowSelf vs. OpenAI-O1 and DeepSeek-R1 (Figure 1)

Task: "Put a hot mug in cabinet"

OpenAI-O1 (fails)

Incorrectly tries to take an egg from the microwave after opening it - does not recognize the mug needs heating.

KnowSelf Llama-8B (succeeds)

Uses <Reflection> token to realize it should heat the mug first using the open microwave, then place it in the cabinet.

Task: "Put a saltshaker in drawer"

DeepSeek-R1 (fails)

Reflects that the drawer is open and tries to close it first, but closing fails and blocks progress.

KnowSelf Llama-8B (succeeds)

Applies <Knowledge> token noting that the agent should place the object directly without removing unrelated items.

Key Insight: KnowSelf's situational self-awareness enables appropriate cognitive strategy selection - reflection when useful, knowledge-seeking when necessary - avoiding the failures of over-confident baseline models.

KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using Only 15% Knowledge

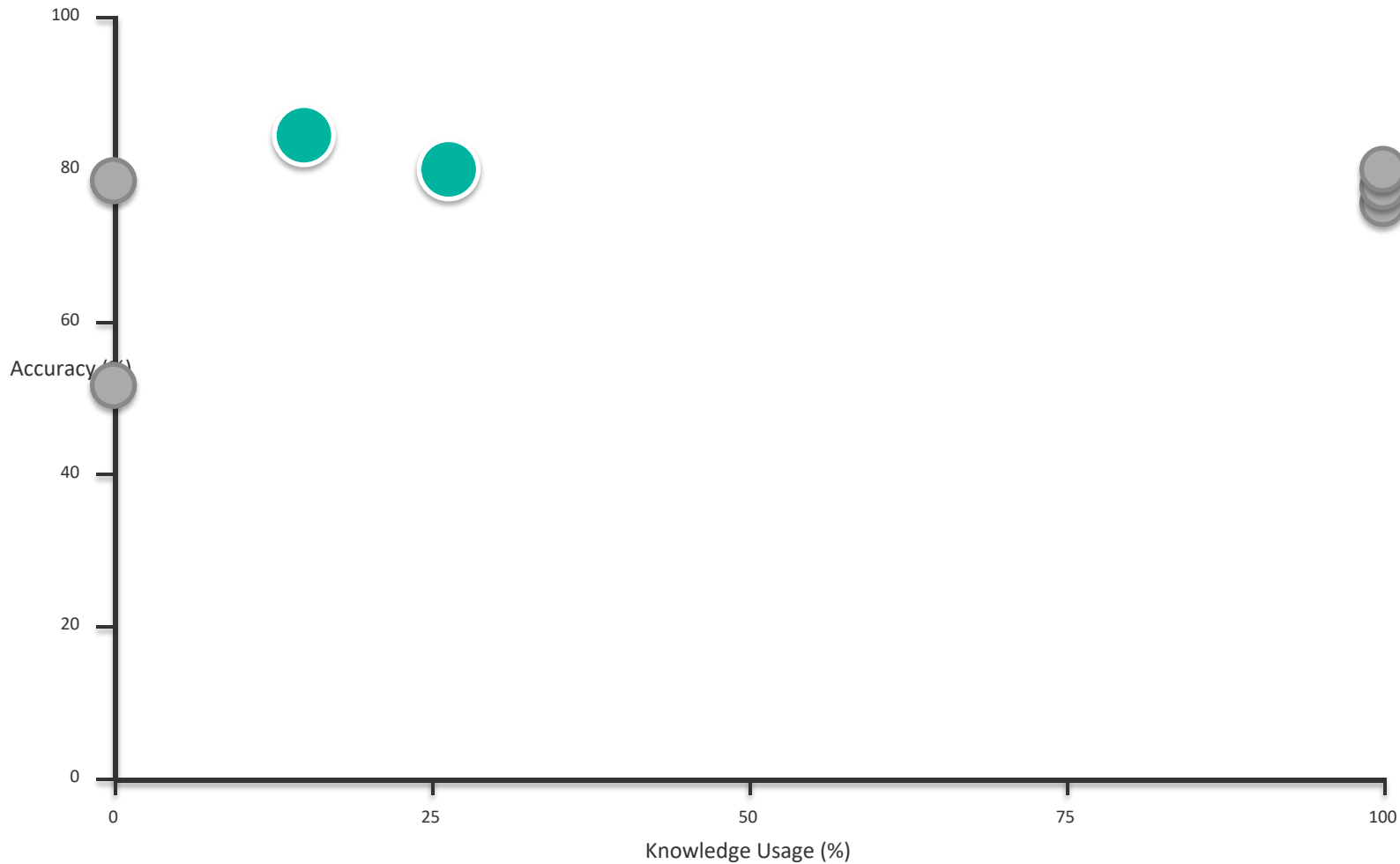
Success rates (%) across six ALFWorld task types - Know% indicates external knowledge usage

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Green highlight = KnowSelf. Best overall scores: Llama-8B KnowSelf (84.33%), Gemma-2B KnowSelf (79.85%)

Minimal Knowledge, Maximum Impact

KnowSelf's efficiency advantage over full-knowledge methods



KnowSelf (selective knowledge)

Full-knowledge methods (100%)

Key Findings:

- KnowSelf uses only 15-26% knowledge vs. 100% for competing methods
- Yet achieves best overall accuracy (84.33% with Llama-8B)
- Surpasses GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%)
- Even Gemma-2B reaches 79.85% matching GPT-4o ExpeL
- Scaling: smaller models benefit significantly from self-awareness

Key Takeaways: Self-Awareness Is the Missing Piece

Five key findings from KnowSelf research

1

Indiscriminate knowledge injection is wasteful and suboptimal

Methods using 100% knowledge (ExpeL, KnowAgent, WKM) often underperform selective approaches that use only 15-26%.

2

Agents can learn situational self-awareness via special tokens

Lightweight heuristic labeling on self-explored trajectories enables self-awareness without expensive human annotation.

3

KnowSelf with Llama-8B outperforms strong baselines

84.33% vs GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%) - demonstrating that smaller models with self-awareness beat larger models without it.

4

Only 15-26% knowledge usage needed vs. 100% for competing methods

KnowSelf proves that selective knowledge application is more effective than blanket knowledge injection.

5

The paradigm generalizes across model scales

Gemma-2B with KnowSelf (79.85%) matches GPT-4o ExpeL, showing the approach scales down effectively.

Future Directions: Extending the three-situation framework to multi-modal tasks, improving knowledge retrieval accuracy, and applying self-awareness to real-world agent deployments.